AI-Powered Healthcare Risk Prediction System

Predicting patient stroke risk and explaining the factors behind it

Dataset: Kaggle Healthcare Stroke Prediction | 5,109 patients | binary classification
Deployed model: **K-Nearest Neighbors** | Benchmarks: Random Forest, XGBoost

Technical & Business Report

Executive Summary

Hospitals triage patients with limited resources and inconsistent manual risk assessment. This project delivers a reproducible, explainable machine-learning system that scores each patient's stroke risk, sorts patients into clinical-action bands, and explains every prediction so clinicians can trust and act on it.

- **Problem:** identify high-risk stroke patients early, despite a severely imbalanced dataset (only 4.9% of patients had a stroke).
- **Solution:** an end-to-end pipeline — cleaning, feature engineering, imbalance handling, three tuned models, SHAP explainability, and a risk-stratification layer that maps probability to clinical action.
- **Deployed model:** K-Nearest Neighbors, tuned and operated at an **80% recall** point — it catches 4 of every 5 strokes.
- **Business value:** earlier intervention, better patient prioritization, and transparent, auditable decisions that support (not replace) clinicians.

Metric	Value
Patients (after cleaning)	5,109
Stroke prevalence	4.87% (249 cases)
Deployed model	K-Nearest Neighbors (k=21)
Operating recall	80% (threshold 0.286)
Test ROC-AUC (KNN)	0.751
High-risk patients flagged	2,135 (41.8%)

1. The Dataset

The Kaggle *Healthcare Stroke Prediction* dataset: 5,110 patient records (5,109 after removing one invalid row), each labelled with whether the patient had a stroke. It mixes demographic, lifestyle, and clinical features.

Feature	Description	Type
age	Patient age in years	Numeric
gender	Male / Female	Categorical
hypertension	High blood pressure (0/1)	Binary
heart_disease	Heart disease (0/1)	Binary
ever_married	Marital status	Categorical
work_type	Employment type	Categorical
Residence_type	Urban / Rural	Categorical
avg_glucose_level	Average blood glucose	Numeric
bmi	Body Mass Index	Numeric
smoking_status	Smoker / former / never / unknown	Categorical
stroke	TARGET — had a stroke (0/1)	Binary

The central challenge is class imbalance: only about 1 patient in 20 had a stroke. Plain accuracy is therefore misleading — a model that predicts 'no stroke' for everyone scores ~95% while catching zero strokes. The whole project is built around this fact.

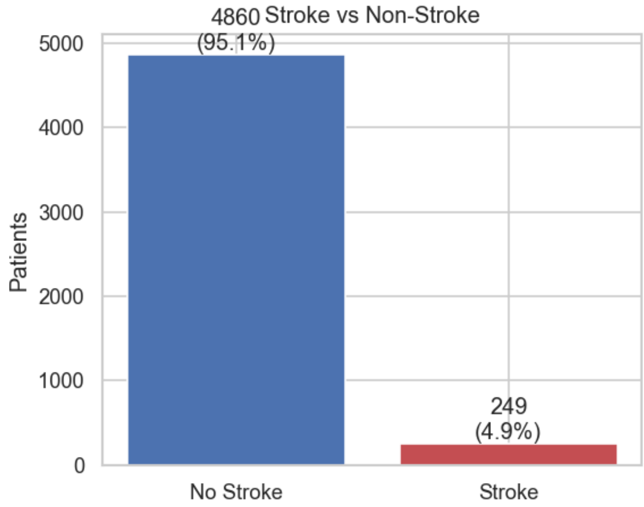

Figure 1. Stroke vs non-stroke — the 95/5 imbalance.

2. Data Cleaning & Feature Engineering

Cleaning (Phase 2):

- **Missing BMI:** 201 values stored as 'N/A' → converted to numeric and imputed with the median.
- **Dropped** the non-predictive *id* column and one record with gender 'Other' (a singleton category that cannot be modelled).
- **Duplicate removal** and categorical text standardization (consistent casing).
- **Outlier handling:** BMI and glucose winsorized at the 0.5 / 99.5 percentile so a few extreme values don't distort scaling.

Engineered features (Phase 4):

- **Age groups:** Child / Young / Adult / Senior.
- **BMI categories:** Underweight / Normal / Overweight / Obese (WHO bands).
- **Glucose risk categories:** Normal / Moderate / High.
- **Composite Health-Risk Score (0–8):** an interpretable additive score combining age, BMI, hypertension, and heart disease.

These features make the model's reasoning clinically legible and gave the health-risk score real predictive weight (see Section 6).

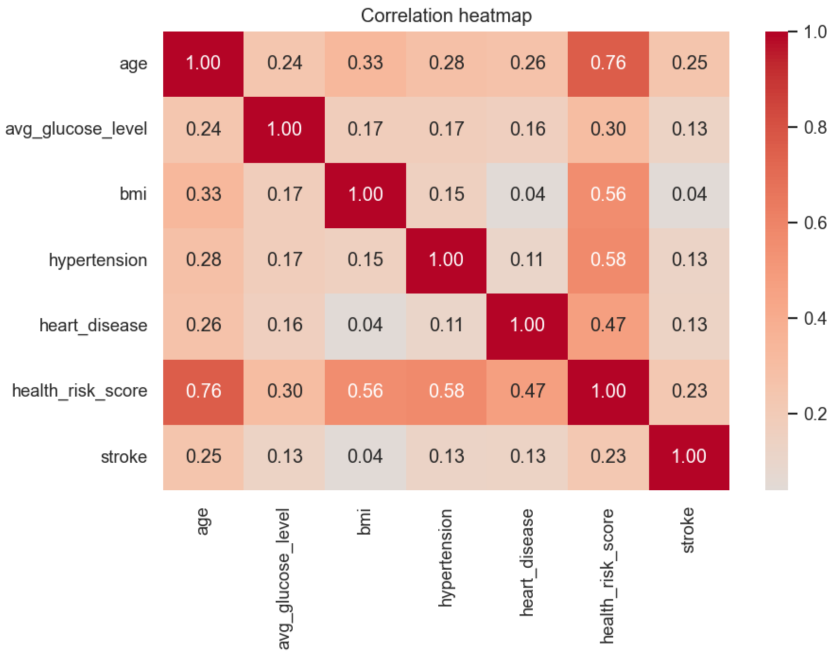

Figure 2. Correlation heatmap — age shows the strongest association with stroke.

3. Exploratory Data Analysis

Three data-understanding questions framed the EDA:

- **How many patients have stroke?** 249 of 5,109 (4.87%) — severe imbalance.
- **What age groups are most affected?** Risk rises steeply with age; Seniors (65+) dominate the positive class.
- **Which health factors matter most?** Age first, then glucose, BMI, hypertension, and heart disease.

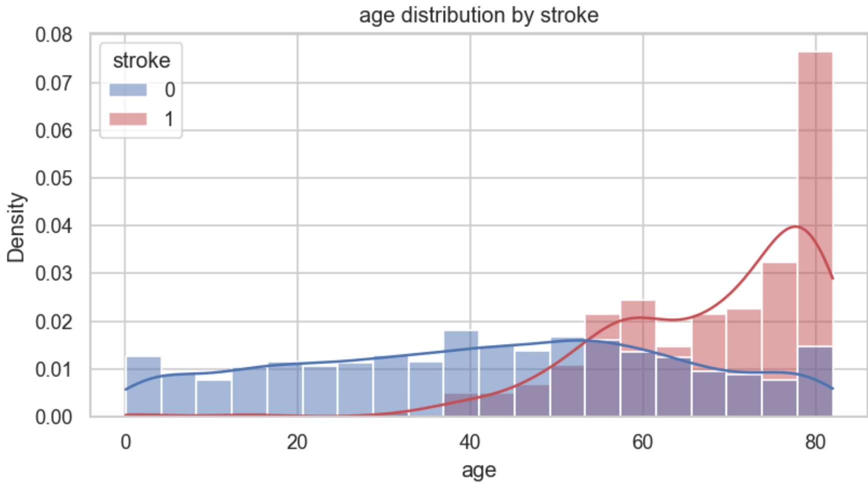

Figure 3. Age distribution by stroke status — stroke patients skew much older.

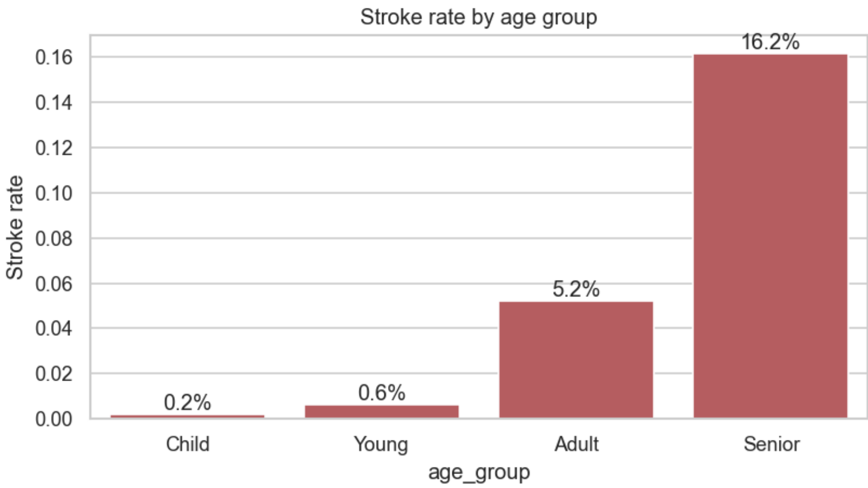

Figure 4. Stroke rate climbs sharply from younger to senior age groups.

4. Class Imbalance Handling

Three standard strategies were compared on a fixed base XGBoost (test set). SMOTE and ADASYN synthesize minority examples; class-weighting penalizes minority errors instead.

Strategy	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
SMOTE	0.9413	0.2727	0.12	0.1667	0.7718	0.1703
ADASYN	0.9393	0.2692	0.14	0.1842	0.7841	0.1856
ClassWeighting	0.8513	0.1507	0.44	0.2245	0.7869	0.1554

Takeaway: class-weighting produced by far the best recall (0.44 vs ~0.13) at comparable ROC-AUC — confirming that for this problem, how we handle imbalance matters more than raw accuracy. SMOTE is used inside the model pipelines, applied *only* within cross-validation folds to avoid data leakage.

5. Models & Hyperparameter Tuning

Three classifiers were tuned with Grid / Randomized search over Stratified 5-fold cross-validation, scored on ROC-AUC:

K-Nearest Neighbors (deployed) — Grid Search

Parameter	Best value
n_neighbors	21
weights	uniform
metric	euclidean

Random Forest — Randomized Search

Parameter	Best value
n_estimators	200
max_depth	6
min_samples_split	5
min_samples_leaf	1
criterion	gini

XGBoost — Randomized Search

Parameter	Best value
n_estimators	200
max_depth	3
learning_rate	0.05
subsample	0.8
colsample_bytree	0.9

6. Results

Test-set performance (default 0.5 threshold):

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	CV ROC-A
knn	0.7025	0.1152	0.76	0.2	0.7507	0.1261	0.7643
random_forest	0.7603	0.1375	0.74	0.232	0.8192	0.2102	0.8234
xgboost	0.6595	0.1016	0.76	0.1792	0.7717	0.1821	0.8087

KNN is the deployed model (ROC-AUC 0.751). Random Forest is the highest-AUC benchmark (ROC-AUC 0.8192); it is kept in the comparison for transparency. KNN was chosen deliberately: it is inherently interpretable (a prediction is the outcome of the most similar past patients), needs no distributional assumptions, and its tuned recall is competitive. The AUC gap is small relative to this dataset's inherent ceiling (~0.82).

Recall-priority operating point

Healthcare prioritizes **recall** — missing a high-risk patient is the costly error. The decision threshold is tuned to **0.286**, giving recall 80%, precision 9.6%, F1 0.17. The table below shows the tradeoff explicitly:

Target recall	Threshold	Actual recall	Precision	Strokes caught	False alarms	Alarms / catc
0.7	0.524	0.76	0.115	38	292	7.68
0.8	0.286	0.8	0.096	40	376	9.4
0.9	0.0	1.0	0.049	50	972	19.44

Pushing recall from 80% to 90% roughly doubles the false-alarm burden. We adopt ~80% recall (the highest KNN reaches without collapsing to predict-all-positive). A hospital can slide this threshold to match screening capacity.

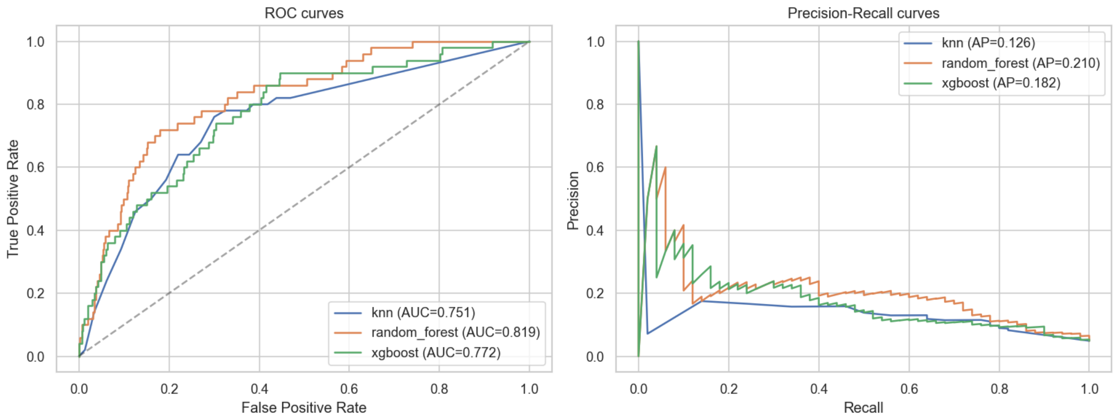

Figure 5. ROC and Precision-Recall curves for all three models.

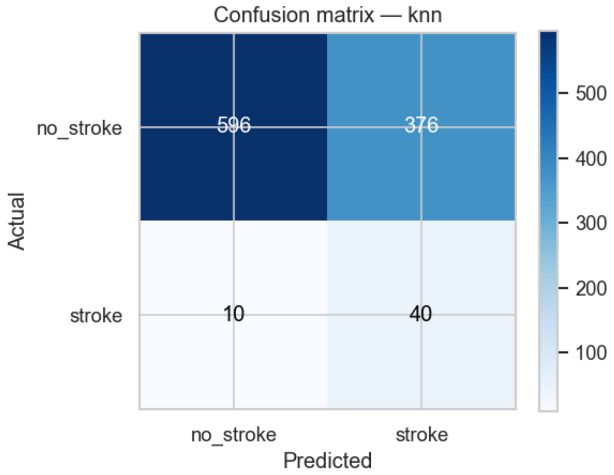

Figure 6. KNN confusion matrix at the deployed threshold (0.286).

7. Explainable AI (SHAP)

Because KNN is not tree-based, the model-agnostic **KernelExplainer** was used to attribute each prediction to its features. SHAP confirms the model reasons from established clinical risk factors — which is what makes it trustworthy to clinicians.

Top global drivers: **age** (dominant), the composite health-risk score, average glucose, BMI, and smoking status.

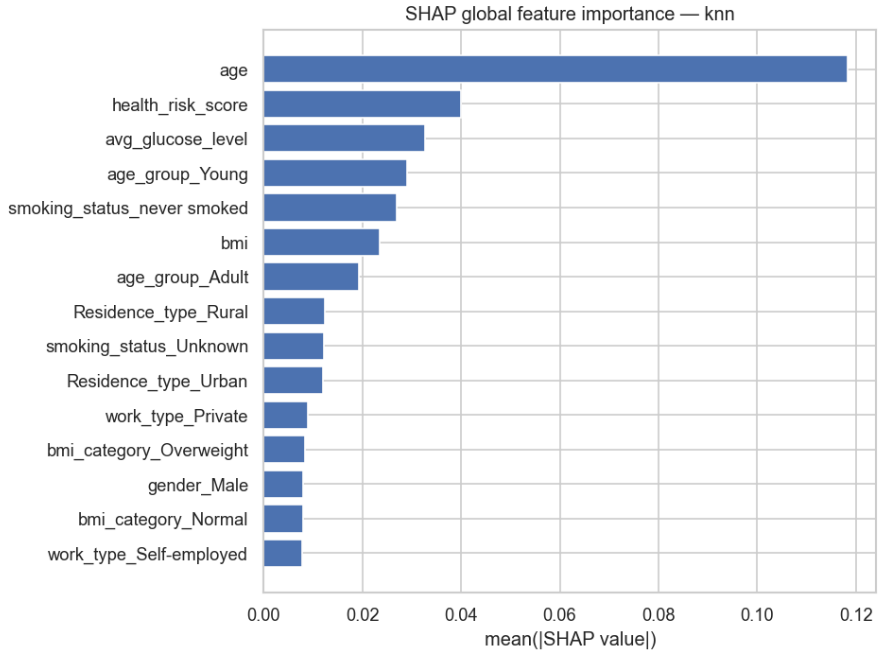

Figure 7. Global SHAP feature importance — age dominates.

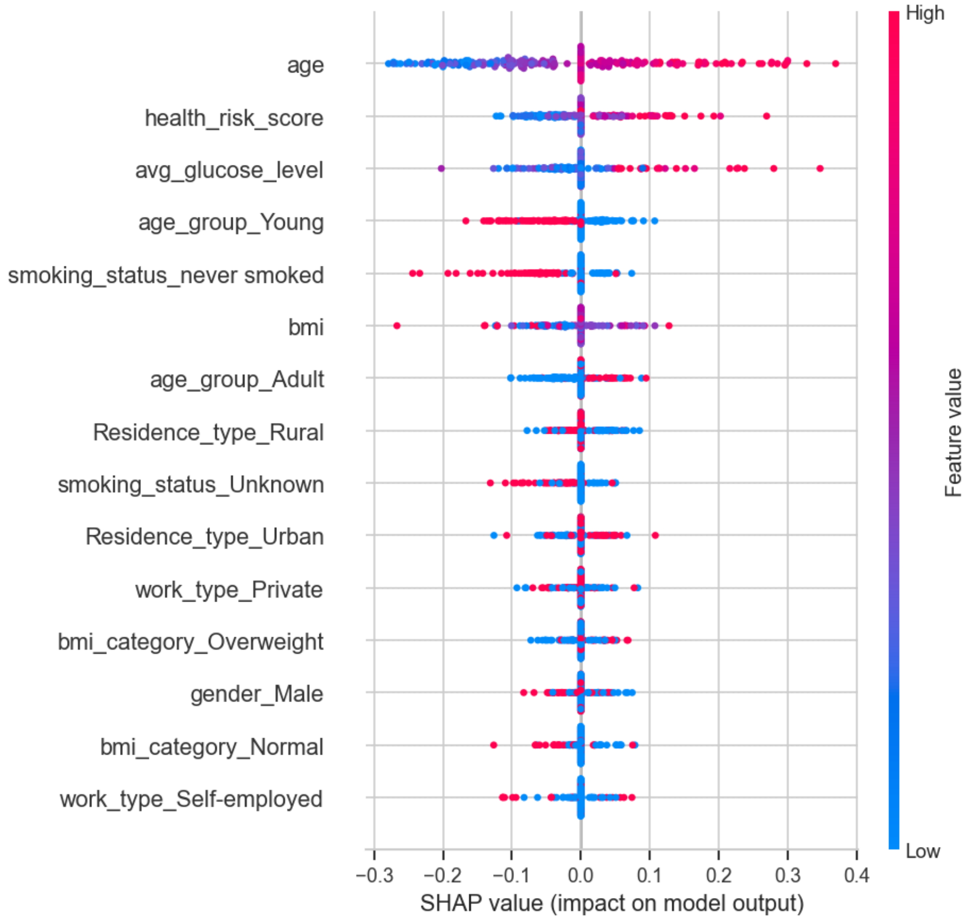

Figure 8. SHAP beeswarm — direction and magnitude of each feature's effect.

Local explanations answer *"why was this patient flagged?"* — a waterfall plot per patient, exactly what a clinician would see beside the score:

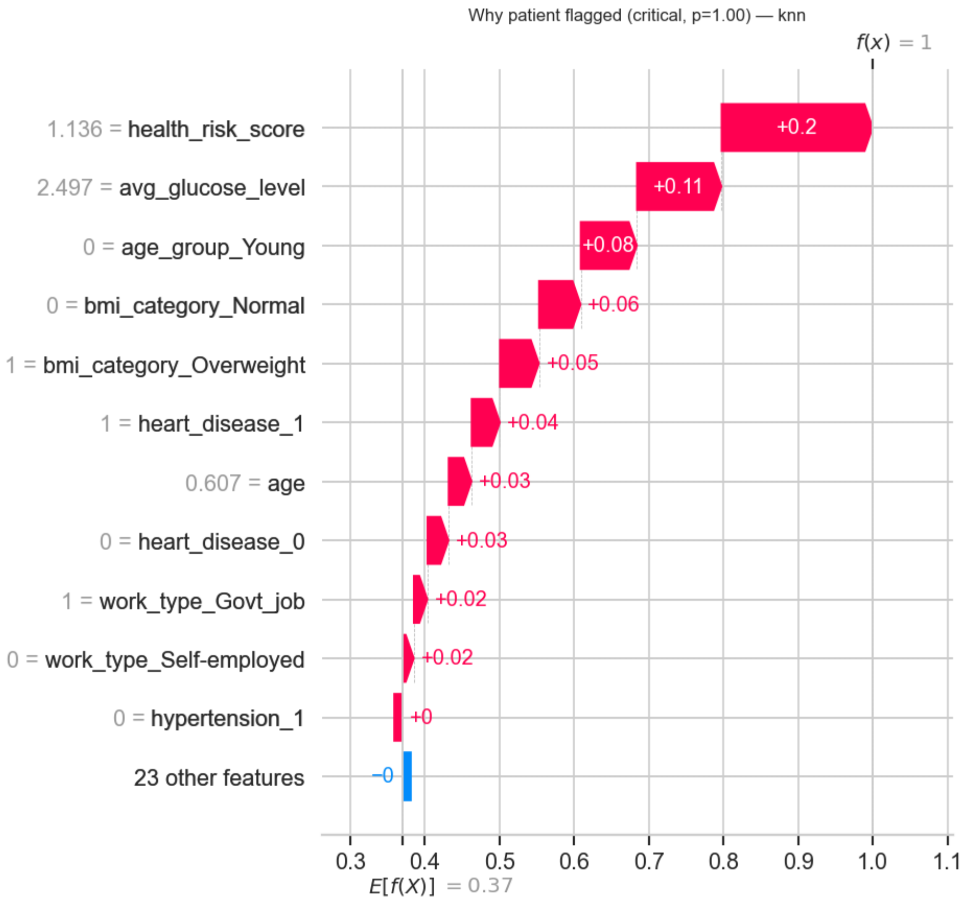

Figure 9. Local explanation for a high-risk (Critical) patient.

8. Risk Stratification → Clinical Action

Model output is converted into four action bands. Every patient is scored with leakage-free out-of-fold probabilities, then assigned a band:

Risk Level	Predicted prob.	Action	Patients	Actual stroke rate
Low	0–10%	Routine Monitoring	2633	1.0%
Medium	10–30%	Follow-up	341	5.9%
High	30–60%	Specialist Review	731	5.9%
Critical	60%+	Immediate Attention	1404	11.4%

The actual stroke rate rises across the bands — lowest in Low (~1%) and highest in Critical (~11%) — confirming the stratification is clinically meaningful and that prioritizing higher bands for review is justified. (Medium and High are close, a known effect of KNN's coarse probabilities.)

9. Business Value

Benefit	Impact
Early Detection	Flags high-risk patients before symptoms escalate — faster intervention.
Resource Optimization	Ranks patients so limited specialist time goes to those who need it most.
Reduced Healthcare Costs	Preventing severe stroke outcomes is far cheaper than treating them.
Clinical Decision Support	Explains every score with SHAP — assists, never replaces, the clinician.
Consistency	Replaces inconsistent manual assessment with a uniform, auditable standard.
Population Health Monitoring	Aggregated dashboards track risk by age, gender, and region.

Deliverables

- Three tuned models (KNN deployed; Random Forest & XGBoost benchmarks), saved.
- Patient-level risk scores with band + recommended clinical action.
- Reports: EDA, model comparison, clinical recommendation (this document).
- Power BI-ready data exports + dashboard build guide (Executive Summary, Risk Analysis, Model Performance pages).

Honest limitations

- ROC-AUC ~0.75–0.82 is near this dataset's realistic ceiling — the features carry limited signal. Any '95%+ accuracy' claim simply predicts 'no stroke'.
- KNN emits coarse, stepped probabilities (k=21), making its risk bands blockier than a tree model's — documented rather than hidden.
- This is a decision-support tool, not a diagnostic device; all flags require clinical confirmation.